Hertie School

Beyond the Ballot: TikTok Virality and Political Engagement in Nepal's 2022 Elections

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Abstract

Nepal's 2022 elections marked a turning point in political communication, with social media platforms such as Facebook, YouTube, and TikTok emerging as powerful arenas for political engagement amid widespread youth disillusionment with traditional parties. Beyond platform adoption, virality on TikTok—measured through likes, shares, and comments—became an important part of how political narratives circulated and gained visibility online.

Despite TikTok's growing influence, little research has systematically examined how political content gains visibility on the platform, particularly in low-income, multilingual democracies such as Nepal. Understanding virality in this context is methodologically innovative—leveraging multimodal machine learning on audio, image, and text data—and substantively important for explaining how digital platforms reshape electoral dynamics.

This study analyzes 2,964 political TikTok videos accessed via TikTok's Research API, applying machine learning and regression-based association analysis to examine patterns of engagement. A leakage-safe virality prediction pipeline shows that political virality is moderately predictable from pre-upload information, but that sender characteristics dominate performance. Across alternative target definitions, models that include sender variables outperform content-only and platform-only setups, indicating that structural creator capacity matters more than multimodal content features alone.

For content-level analysis, regression results show that videos associated with independent political actors achieve higher engagement on average than general election-themed or legacy-party content, even after adjusting for follower reach. By contrast, communication-style coefficients are unstable across model specifications and diagnostic checks. Propensity-score and within-creator diagnostics indicate that styles such as Charisma/Leadership are highly endogenous to specific actors, limiting causal interpretation and suggesting that styles operate as structural signals rather than transferable tactics.

Taken together, the findings support a framework in which political virality on TikTok is shaped primarily by sender capacity, secondarily by actor-linked content signaling, and only indirectly by platform affordances observed through

metadata. The study contributes an open-source multimodal analysis pipeline, an explicit discussion of operationalization tradeoffs, and a clearer framework for studying political virality on short-video platforms without overstating causal claims. In plain terms, the policy lesson is simple: social media can widen access to political information and help newer actors gain attention, but election governance on short-video platforms should focus not only on harmful content, but also on who gets seen, how that visibility can be checked, and whether public institutions in Nepal can independently audit political reach.

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List of Abbreviations

Abbreviation	Definition
API	Application Programming Interface
CPN UML	Communist Party of Nepal (Unified Marxist-Leninist)
LaBSE	Language-agnostic BERT Sentence Embedding
MiniLM	Mini Language Model
OLS	Ordinary Least Squares
SHAP	SHapley Additive exPlanations
TikTok	Social media platform for short-form videos
XGBoost	eXtreme Gradient Boosting

1. Introduction

Nepal's 2022 local elections (2079 B.S.) marked a visible shift in political communication, with TikTok joining Facebook and YouTube as an important campaign arena (Nepali Times, 2025). Political actors used these platforms to circulate speeches, campaign clips, satire, music, and commentary, especially among younger voters who were increasingly frustrated with legacy parties (Dahal, 2023; Explainers, 2024). In this sense, election communication was no longer shaped only by party organizations, news outlets, and rallies. It was also shaped by short-video platforms and the creators who could attract attention on them.

This shift mattered especially for independent candidates. Balendra Shah (Balen Shah), for example, became highly visible online through a mix of campaign symbolism, youth appeal, and creator support (Deutsche Welle, 2022). The "lauro" (stick) hashtag became a recognizable symbol of the independent movement (The Annapurna Express, 2022), and influential creators such as "Routine of Nepal Banda" helped increase the visibility of newer political actors (Dahal, 2023). Balen Shah ultimately won the Kathmandu mayoral race with 38.6% of the vote (Wikipedia contributors, 2022). Legacy parties remained dominant overall, but the campaign showed that newer actors could use social media to gain public attention outside traditional party and media structures.

TikTok matters analytically because it can circulate content beyond a creator's immediate follower network and because it makes remix, sound, and visual performance central to political communication (Guinaudeau et al., 2022). At the same time, the platform's internal ranking system is not directly observable to researchers. This means that TikTok is important to study, but it must be studied carefully: we can observe outcomes and metadata, but not the platform's full recommendation process. In Nepal, this matters even more because TikTok became politically controversial after it was banned in 2023 and reinstated in 2024 amid debates about misinformation, governance, and digital regulation (Lamichhane, 2024; Reuters, 2024).

Despite growing API access (TikTok, 2025), little research has examined political virality on TikTok in Nepal through a multimodal lens. This thesis addresses that gap by asking two simpler questions. First, how well can virality be predicted from pre-upload sender, content, and platform features? Second, how are communication styles and political content themes associated with virality? The goal is not to infer TikTok's hidden algorithm, but to explain observable patterns of political attention during an election period.

This study offers four key contributions:

- **Academic**: It examines a substantively important but under-studied case of election-related TikTok communication in Nepal, contributing evidence from a multilingual Global South context.
- **Methodological**: It develops and evaluates a multimodal pipeline for virality prediction while explicitly testing how results change when the virality target is defined in more or less view-dependent ways.

- **Conceptual**: It proposes a framework in which political virality is shaped by sender capacity, content signaling, and platform affordances, while emphasizing that observed style effects are actor-bound associations rather than proven causal strategies.

- **Policy-facing**: It suggests that, in Nepal, the Election Commission and other digital-policy actors should protect the civic benefits of social media while improving transparency of content distribution, research access to election-period data, and recognition of digital reach as a politically relevant campaign resource.

This research lays the groundwork for future cross-platform virality studies (e.g., TikTok, YouTube Shorts) and for more tightly identified work on political visibility in low-income electoral contexts.

2. Literature Review

2.0.1 Social Media, Political Attention, and Algorithmic Gatekeeping

Social media has changed how political attention is distributed. Earlier agenda-setting research focused on editors, journalists, and party elites as the main gatekeepers of public visibility. On digital platforms, ranking systems and user engagement now play a larger role in deciding which messages travel widely and which do not.

Evidence from (Gilardi et al., 2022) suggests that digital platforms have changed agenda-setting power without eliminating older actors entirely. Gilardi et al. show that media agendas, party agendas, and politician agendas now shape one another rather than following a single hierarchy. For this thesis, the key point is simpler: visibility is increasingly shaped by platform systems as well as by political actors themselves.

In Nepal, social media has become an important space for political expression, especially among younger citizens who are dissatisfied with legacy parties and traditional media (Bhandari, 2024; Dahal, 2023). Yet the mechanisms behind online political visibility remain underexplained. Existing work is often descriptive, leaving open the question of why some political content receives much more attention than other content during elections.

2.0.2 Conceptualizing Virality as Engagement-Weighted Political Attention

"Virality" is widely used but not always clearly defined. (Berger & Milkman, 2012) note that the term is often used loosely to refer either to reach or to speed of diffusion. On platforms such as TikTok and YouTube, many studies reduce virality to view counts alone (Nissenbaum & Shifman, 2017). That is convenient, but it also mixes together two different things: how much a video circulated and

how much baseline visibility the sender already had.

Recent computational work argues that virality should also reflect active engagement, not just passive exposure. (Sah & Jordan, 2025b) treat comments, shares, and reposts as stronger signals of attention than views alone. This matters in politics, where provocative or emotionally resonant content may generate strong reactions even when it does not come from the largest accounts.

Following this literature, the thesis treats political virality on TikTok as an engagement-weighted attention outcome. The goal is not to claim that this is the only valid measure of virality, but to use a measure that captures both visibility and user response while reducing reliance on raw view counts alone.

2.0.3 Affective Information Behavior and Political Engagement in Nepal

Research in political psychology increasingly treats emotion as part of political reasoning rather than as its opposite. This is relevant to TikTok because short-video platforms reward attention, reaction, and rapid interpretation.

Affective Intelligence Theory (AIT) offers one useful way to think about this relationship (Marcus et al., 2013). In simple terms, the theory argues that different emotions push people toward different kinds of political processing. Enthusiasm can reinforce existing attachments, while anxiety can interrupt routine thinking and make people pay closer attention to new information. Anger, by contrast, can deepen existing loyalties and sharpen group boundaries. For this thesis, the value of AIT is not that it proves a platform effect, but that it helps explain why emotional tone may matter for political attention online.

This perspective also fits recent work from Nepal. (Gurung & Gnawali, 2025) analyzed Facebook emoji reactions during the 2022 Kathmandu mayoral race and found that emotional engagement varied strongly across candidates. Independent candidate Balen Shah drew unusually strong positive engagement, while establishment candidates more often attracted negative or mocking reactions. The result does not prove that online emotion determines electoral outcomes, but it does suggest that digital engagement can provide one useful signal of political attention and sentiment during campaigns.

2.0.4 Platform Logic, Affordances, and Content Amplification

Political visibility online is also shaped by platform affordances: the design choices and ranking systems that affect what users see and engage with. As (Naranjo-Vinueza et al., 2025) note, these affordances are not fully deterministic, but they do shape the conditions under which attention is distributed.

Interest graph vs. social graph: Unlike follower-centered platforms, TikTok can recommend content beyond an account's immediate network (Kristi Lang, 2025; The Shelf, 2025). This is one reason why it matters politically: some content may travel widely even when it does not come from the largest accounts.

Emotional content and engagement: Prior research suggests that engagement-based systems often reward emotionally charged content. (Brady et al., 2017) show that moral-emotional language on Twitter increased retweet rates, while (Milli et al., 2025) find that engagement-based ranking can amplify anger, sadness, and anxiety in political content.

Feedback dynamics: When emotionally charged content receives stronger reactions, those reactions can become further visibility signals. This does not mean every emotional message goes viral, but it does suggest a plausible mechanism through which emotional and confrontational political content can receive disproportionate attention.

2.0.5 Integrating the Framework: From Theory to Modeling Decisions

Taken together, this literature suggests a simple way to think about virality in Nepal's 2022 election period. Virality is better understood as platform-mediated political attention than as a direct measure of persuasion or electoral impact.

Three ideas matter most for this thesis. First, **sender capacity** matters because videos do not enter TikTok on equal footing. Prior audience size, creator history, and account credibility shape how much attention a video can attract from the start. Second, **content signaling** matters because political videos carry style, identity, and emotional tone, not just information. Third, **platform affordances** matter because timing, ranking, remix features, and recommendation systems shape the environment in which sender and content signals are received.

The framework is intentionally modest. Sender and content factors can be observed directly in the data, while platform dynamics are only partly visible through metadata. This is why RQ1 focuses on prediction from observable pre-upload features, and why RQ2 is limited to conditional associations rather than causal claims about communication style.

3. Conceptual framework

This study treats political virality on TikTok as a form of platform-mediated political attention shaped by three elements: sender capacity, content signaling, and platform affordances (Figure 3.1). Rather than reducing virality to reach or view counts alone, the framework emphasizes that visibility and engagement are shaped by both structural and contextual factors. Platform dynamics are included, but only indirectly, because the platform's internal recommendation process is not observable in the available data.

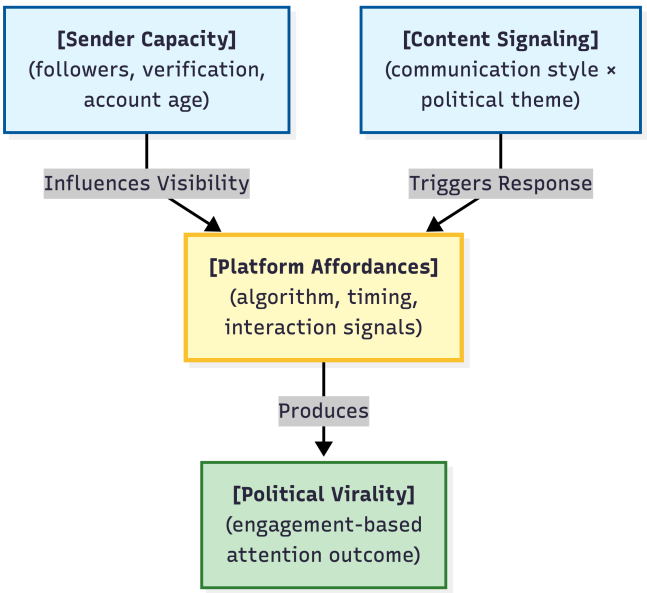

Figure 3.1: Conceptual Framework

Sender capacity captures the structural advantages that shape how content enters the platform environment. Follower count, prior creator engagement, verification status, and account tenure are treated as core predictors and controls because they proxy baseline audience reach, creator prominence, and accumulated platform familiarity. In this framework, sender capacity is expected to explain an important share of virality even before multimodal content features are considered.

Content signaling refers to how political messages are framed and performed, including communication styles (e.g., Charisma/Leadership, Comedic, Critique/Frustration) and political content themes (e.g., independent candidates versus legacy parties). In this thesis, styles are treated not as freely transferable tactics but as signals that may reflect identity, authenticity, and emotional orientation. Their association with engagement is therefore expected to depend partly on who is speaking and in what political context.

Platform affordances encompass TikTok's ranking environment and temporal dynamics, such as posting time and proximity to election day. Because the platform's internal recommendation process is unobserved, this study does not identify algorithmic effects directly. Instead, platform affordances are approximated through observable metadata and treated as part of the context within which sender and content factors operate. Within this framework, virality is defined as engagement-weighted political attention rather than raw popularity.

Together, these components inform the study's modeling strategy. The composite virality index reflects engagement-based attention; sender and timing variables capture structural advantages; and style and theme variables describe how political content is performed. The framework also clarifies the limits of causal inference: because communication styles are tied to specific actors and do not have strong counterfactual overlap in this dataset, their coefficients are interpreted as conditional associations rather than causal treatments.

4. Research Questions

The study addresses two research questions (RQs):

RQ1: **How well can political TikTok virality be predicted from pre-upload sender, content, and platform characteristics, and which feature groups contribute most to that prediction?**

RQ2: **How are different communication styles and political content themes associated with virality?**

5. Research Design

This study employs a mixed-methods approach, combining machine learning for virality prediction (RQ1) and regression-based content analysis for style–theme interactions (RQ2), to examine TikTok's role in Nepal's 2022 local elections. The design leverages multimodal data (text, audio, visual) and platform metadata, integrating computational techniques with political communication theory.

5.1 Data Sources and Collection Strategy

The dataset comprises 28,165 TikTok videos collected from March 15, 2022 (one month before candidacy registration) to May 13, 2022 (election day), capturing the campaign period of Nepal's 2022 local elections. A hybrid approach was used, combining the Official TikTok Research API, the Unofficial TikTok-Api, and a custom subclass of TikTok-Content-Scraper (Bukold, 2025). The Official API collected video metadata (e.g., views, likes, shares, hashtags, descriptions) and user metadata (e.g., follower count, verified status) but was limited by quotas (1,000 users/day, 100,000 videos/day). The Unofficial API supplemented user-level data, while the custom scraper captured platform-specific features like `video_diversification` labels, duet/stitch behavior, and music metadata.

Videos were collected via hashtag-based searches (e.g., `#NepalElection2022`, `#चुनाव`, `#LocalElection2022`), adapted from (Pinto et al., 2024) (see Appendix 9). After removing duplicates and irrelevant content, the final dataset included 28,165 videos. Initial features collected include:

- **Platform/Engagement Metrics**: Views, likes, shares, comments.
- **User Features**: Follower count, video count, verified status.
- **Content Metadata**: Hashtags, video descriptions, audio metadata.
- **Temporal Features**: Posting timestamps (UTC).

The data collection pipeline and schema are illustrated in Figures 5.1 and 5.2, respectively.

5.1.1 Sampling

For multimodal analysis, a stratified sample of 3,000 videos was drawn from the 28,165 videos based on view count percentiles:

- **High-Virality**: $\geq$ 95th percentile ($\geq$ 8,000 views; $\sim$5% of dataset).
- **Medium-Virality**: 60th to 95th percentile ($\sim$35% of dataset).
- **Low-Virality**: $\leq$ 60th percentile ($\sim$60% of dataset).

After removing videos that became private or irrelevant, the final sample was 2,964 videos.

As illustrated in Figure 5.2, the dataset is further enriched post-sampling with audio transcripts and image analysis. These additional features, derived through the processing pipeline depicted in Figure 5.1, complete the multimodal data collection for the sampled dataset (detailed in the following section).

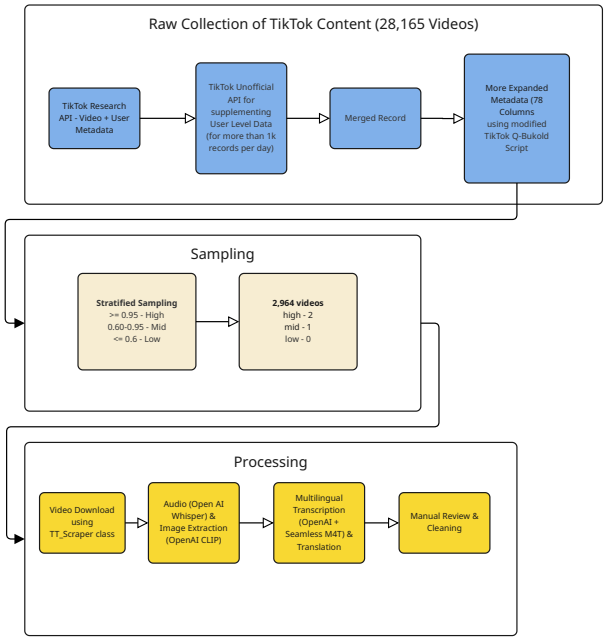

Figure 5.1: Data Collection Pipeline

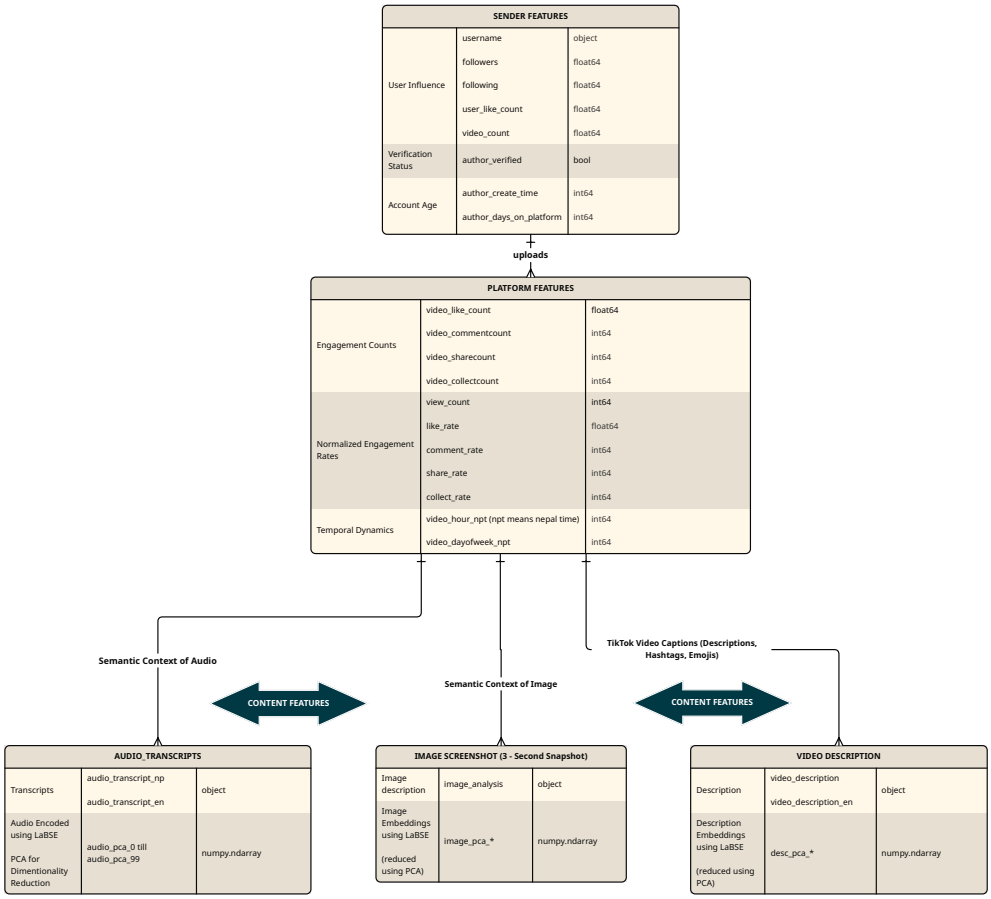

Figure 5.2: Data Schema Collection

5.2 Feature Engineering

5.2.1 Operationalizing Virality (Dependent Variable)

While there is no universally accepted standard for measuring virality on TikTok, several research studies and commercial tools have proposed their own metrics and predictive models (Roring, 2024; Sah & Jordan, 2025a). Moreover, no prior research has applied standard virality criteria to TikTok videos in the context of Nepal's 2022 elections, necessitating a flexible approach to capture its role in shaping public attention. In this study, virality is operationalized as a composite score emphasizing relative audience engagement, prioritizing interaction over raw popularity. Four **normalized engagement rates** - like, comment, share, repost, were calculated.

The rates were combined with the logarithm of raw view count to form a **virality score**, using a weighted formula that emphasizes active user engagement. The final formula is:

$$\begin{aligned}\text{virality score} = 0.35 \cdot \log(1 + \text{views}) + 0.25 \cdot \log(1 + \text{like rate}) + 0.20 \cdot \\ \log(1 + \text{comment rate}) + 0.15 \cdot \log(1 + \text{share rate}) \\ + 0.05 \cdot \log(1 + \text{collect rate})\end{aligned} \tag{5.1}$$

Weightings of (0.35, 0.25, 0.20, 0.15, 0.05) were assigned to prioritize active engagement behaviors. This operationalization is necessarily imperfect: the inclusion of views risks conflating engagement intensity with baseline exposure, while any fixed weighting scheme remains partly researcher-imposed. To address this concern, RQ1 supplements the legacy score with alternative, less view-dependent target definitions as robustness checks. Given the exploratory nature of this study and the absence of an established benchmark for Nepal's TikTok context, the composite score should be understood as an analytically useful proxy for political attention rather than as a definitive measure of virality. The score was transformed into a ternary classification label for RQ1:

- **Low Virality (0)**: Bottom third of scores.
- **Moderate Virality (1)**: Middle third.
- **High Virality (2)**: Top third.

For RQ2, a log-transformed version of the virality score was used as a continuous regression target to account for its skewed distribution.

5.3 Multimodal Feature Extraction

This study extracted multi-modal features from a stratified sample of 2,964 TikTok videos to predict virality. Features were derived from three primary modalities: audio transcripts, video descriptions, and visual frames.

5.3.1 Textual Features

Textual or Content features aim to capture the semantic, emotional, and contextual characteristics of a video's actual media content. These include both linguistic and visual elements:

Audio Transcripts

- Transcribed using OpenAI's GPT-4o Whisper API (OpenAI, 2024), primarily in Nepali with minor instances of Hindi, Bhojpuri/Maithili, Urdu, and English. Manual corrections added contextual annotations (e.g., `{{Party_Name}} Song`). Transcripts were embedded using the BGE-m3 model (Chen et al., 2024) and reduced via PCA (`audio_pca_0` to `audio_pca_99`).

Video Descriptions

- **Video Descriptions**: Language detection via LangDetect (Nakatani, 2014) identified 494 English and 2,470 non-English (Romanized Nepali-English, Devanagari) descriptions. Non-English descriptions were translated to English using GPT-4o (OpenAI, 2024), preserving hashtags and mentions. Descriptions were embedded with BGE-m3 and reduced via PCA (`desc_pca_*`).

```
prompt = f"Translate the following text to English but keep all
hashtags and mentions (like #nepal, @username)
unchanged:\n\n{masked_text}"
```

5.3.2 Visual Features

- **Visual Features**: A frame at the 3-second mark was extracted (TikTok, 2023), captioned using GPT-4o Vision API (OpenAI, 2024), embedded with BGE-m3, and reduced via PCA (`image_pca_*`).
- **Sentiment Scores**: Extracted for audio, descriptions, and image captions using a pretrained sentiment model.
- **Lexical Features**: Included `audio_word_count`, `image_word_count`, and `hashtag_count`.

5.3.3 Platform Features

Platform features refer to platform-mediated affordances and behavioral traces that reflect how users engage with the content:

- **Temporal Dynamics**: `video_hour_npt`, `video_dayofweek_npt`.
- **Metadata**: `video_diversification` labels, duet/stitch indicators.

5.3.4 Sender Features

Sender features capture the video uploader's social capital and credibility:

- **Creator Tier Metrics**: A categorical feature, `creator_tier`, was derived from follower counts to analyze non-linear effects of audience reach on virality. Using the empirical distribution of follower counts across 2,964 videos (25th percentile: ≈ 983, 50th: $\approx 2,191$, 75th: $\approx 8,120$, 90th: $\approx 48,740$, 95th: $\approx 108,700$, 99th: $\approx 376,500$), we defined tiers as: Regular ($< 10,000$ followers), Micro Influencer ($10,000$–$49,999$), Macro Influencer ($50,000$–$99,999$), and Mega Influencer ($\geq 100,000$). This data-driven approach ensures relevance to Nepal's 2022 election context and enhances model interpretability.

- **Verification Status**: A binary feature, `author_verified`, indicates official TikTok verification, reflecting institutional presence or public figure status that may influence platform distribution and audience trust.

- **Account Age**: Derived from `author_create_time`, `author_days_on_platform` measures the creator's active duration on TikTok, capturing their tenure and experience, which may affect content strategy and reach.

5.3.5 Feature Integration & Pre-Processing

Predictor variables from platform metadata, sender attributes, and content modalities were integrated into a unified, model-ready dataset. Numerical features (e.g., follower count, interaction rates like `like_rate`, `comment_rate`) were z-score normalized for comparability, while categorical variables (e.g., day-of-week) were one-hot encoded. Multimodal embeddings from audio transcripts, image snapshots, and video descriptions were generated using a multilingual transformer model, reduced via PCA, and used in their compact latent form without further scaling, as they already captured normalized semantic similarity.

To prevent label leakage, post-performance variables (`video_like_count`, `video_commentcount`, `video_sharecount`, and their normalized counterparts `like_rate`, `comment_rate`, `share_rate`, `collect_rate`) were excluded, focusing strictly on pre-upload features like content semantics (text, image, audio) and user metadata (e.g., `creator_tier`, verification status). This structured predictor matrix provided a consistent input format and enabled comparison of content, platform, and sender contributions to virality prediction.

5.4 Sentence Embedding Model Selection

We evaluate four sentence embedding models on Nepali election-related texts:

- LaBSE (Reimers & Gurevych, 2019)
- paraphrase-multilingual-MiniLM-L12-v2 (MiniLM-L12) (Reimers & Gurevych, 2020)
- all-MiniLM-L6-v2 (Reimers & Gurevych, 2021)
- BGE-M3 (Chen et al., 2024)

BGE-M3 demonstrates multi-lingual capabilities supporting 100+ languages with dense/sparse retrieval (Chen et al., 2024). The MiniLM variants offer optimized performance for semantic similarity tasks (Reimers & Gurevych, 2019).

Based on evaluation of sentence-embeddings (See Table 9.8) for 11-pairs of random sentence-text pairs, BGE-m3 was chosen for its more balanced performance in compared to other models.

5.5 Analytical Framework

The analysis addresses the two research questions through two complementary approaches:

5.5.1 Machine Learning for Virality Prediction (RQ1)

- **Model, Training, and Validation**: RQ1 is estimated using a leakage-safe prediction pipeline that compares sender, content, and platform feature groups across repeated stratified cross-validation folds. The main reported target reduces the weight of raw views relative to engagement rates, while more view-dependent and no-view variants are used as robustness checks. Performance is evaluated using accuracy, macro F1-score, and AUC-ROC.
- **Model Interpretation**: Feature-group comparisons and explainability analysis are used to assess whether sender, content, or platform variables contribute most to predictive performance. The aim is not to infer causality, but to identify which observable pre-upload characteristics carry the most predictive information about later virality.

5.5.2 Identification Limits and Causal Feasibility for RQ2

RQ2 asks how communication styles and political content themes are associated with engagement outcomes on TikTok. A causal interpretation would require a much stronger design. Communication styles are not randomly assigned; creators choose them based on who they are, how they position themselves politically, and what kind of audience they expect to reach.

For that reason, a credible causal comparison would require videos that are similar on key pre-treatment characteristics but differ in style. To check whether such a comparison is plausible in this dataset, this study uses propensity score diagnostics. These scores were estimated for focal styles such as *Charisma/Leadership* using only pre-treatment sender and timing covariates, including verification status, follower count, account age, posting time, and distance from election day.

The results show very limited overlap between videos that use these styles and videos that do not. In practical terms, the groups are too different from one another to support a credible "like-for-like" comparison. Even after trimming the sample, videos coded as *Charisma/Leadership* remain structurally distinct from

non-charismatic videos on observable characteristics.

For this reason, the thesis does not interpret RQ2 coefficients as causal effects of style. Instead, the regression results are treated as conditional associations: they describe which styles and themes appear alongside higher or lower engagement after accounting for a limited set of observed controls, but they do not show that changing style alone would change engagement.

5.6 Data Annotation and Validation

A systematic annotation and validation process was used to assess the quality and limitations of the multimodal features and labels used for the RQ1 prediction models and the RQ2 regression analysis on 2,964 TikTok videos from Nepal's 2022 elections.

5.6.1 Audio Transcripts

A 5% sample ($n = 148$) of Nepali audio transcripts (`audio_transcript_np`), converted to PCA embeddings (`audio_pca_*`), was manually validated, achieving 87.2% accuracy by the primary author and a Cohen's Kappa of 0.462 (moderate agreement) with a secondary evaluator (Landis & Koch, 1977). Despite dialectal challenges, embeddings were retained due to their complementary role with image (`image_pca_*`) and description (`desc_pca_*`) embeddings, with `audio_pca_0` showing a mean $|\text{SHAP}| \approx 0.3$.

5.6.2 Video Descriptions

A 5% sample ($n = 148$) of translated video descriptions achieved 98.6% accuracy, preserving hashtags per best practices (Highfield & Leaver, 2016). These were converted to PCA embeddings (`desc_pca_*`) for the XGBoost model.

5.6.3 Image Embeddings

Image features were derived by captioning a frame with GPT-4o Vision and embedding the resulting caption with BGE-m3 before PCA reduction. These image-derived components (`image_pca_*`) were used directly in the XGBoost model without separate manual validation, with `image_pca_2` showing a mean $|\text{SHAP}| \approx 0.1$ for high-virality predictions.

5.6.4 Style and Content Labels for RQ2

Three annotators labeled all 2,964 videos for communication styles (`style_1_label`, e.g., Comedic) and content themes (`content_label`, e.g., party affiliation) per a predefined codebook. A 150-video sample yielded a Cohen's Kappa of 0.622 (substantial agreement) for `style_1_label` (Landis & Koch, 1977), supporting its use in OLS regression. Content labels, derived via zero-shot classification and manually reviewed, lacked formal validation due to resource constraints.

5.6.5 Discussion

Validation confirms reliability, with 98.6% accuracy for video descriptions and Kappa = 0.622 for style labels, though audio transcripts' Kappa = 0.462 reflects linguistic challenges. CLIP embeddings align with computer vision standards, and zero-shot content labels were mitigated by manual review. This ensures dataset suitability for modeling virality and style–theme interactions, though future work could improve audio transcription using advanced multilingual NLP fine-tuned for Nepalese social media communication context.

5.7 Ethical Considerations

Data collection used publicly accessible TikTok content and complied with the platform access methods described in the data section. No personally sensitive information beyond public account-level metadata was intentionally collected or analyzed, and no contact was made with platform users. Because this study relies on observational platform data, the main ethical concerns are measurement error, contextual misinterpretation, and the responsible reporting of public political speech in a low-resource setting. These limitations and potential biases are reported transparently throughout the thesis.

6. Results

6.1 Exploratory Analysis

Exploratory Analysis was conducted on the full dataset. For Dataset summary (see Table 9.1)

6.1.1 Number of Videos Over Time

Figure 6.1 illustrates the volume of TikTok videos posted over this time frame. A noticeable rise in content production is observed as the election approaches, peaking sharply on Election Day (May 13, 2022).

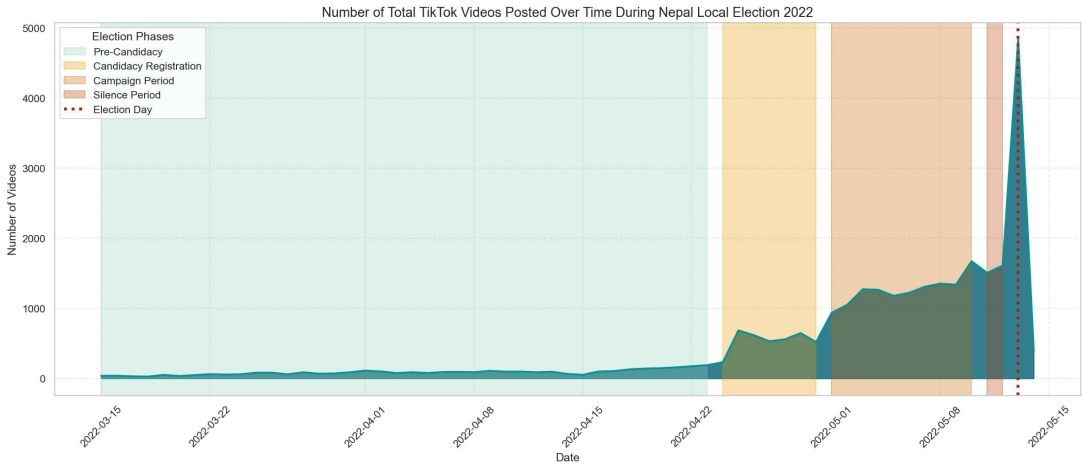

Figure 6.1: TikTok Video Post Over Time)

6.1.2 Post Volume Heat Map

Figure 6.2 reveals that most video posts occur on Fridays, particularly during the late afternoon and evening hours. A distinct posting peak is observed at 21:00 (9:00 PM NPT), suggesting that content creators might be strategically timing their posts to maximize visibility and engagement during the end of their weekdays.

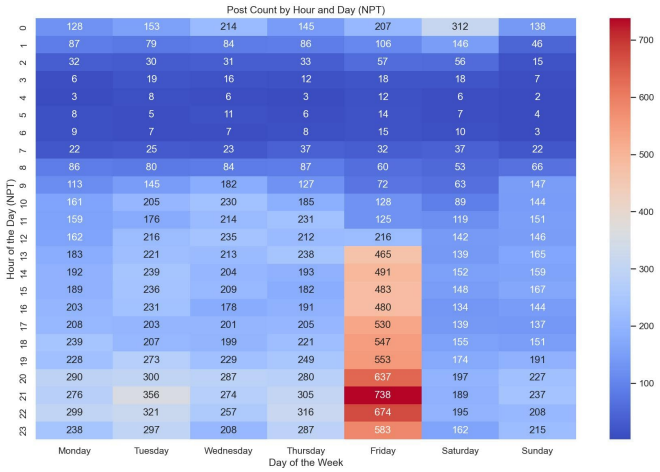

Figure 6.2: TikTok Video Post Over Time)

6.1.3 Top Keywords in Video Descriptions

The dataset includes 20,237 unique hashtags extracted from video descriptions.

After applying a custom stopword filter to remove generic or repetitive terms (e.g., `fyp, viral, trending`), we generated a word cloud to visualize prominent hashtags. Figure 6.4 reveals Nepali keywords like *chunab* ("election") and *chunablagyo* ("election vibes"), as well as direct references to political campaigns. Note that some Devanagari (Nepali) script tokens are rendered as square boxes due to font encoding limitations.

Table 6.3 lists the 20 most frequently occurring hashtags in the political genre along with their respective counts.

Top_20_HashTags_Political_Support

Hashtag	Count
nepalicongress	732
cpnuml	329
congress	324
balen	299
voteforbalen	296
kpoli	286
balenshah	279
jaynepal	240
voteforchange	225
voteforsun	222
mayor	174
yemale	169
uml	167
nekapayemalay	166
prachanda	157
एमाले	123
नेकपा_एमाले_जिन्दाबाद	122
गठबन्धन	114
voteforbalenshah	97
votefortree	92

Figure 6.3: Top Political Hashtags Used During Local Election 2022

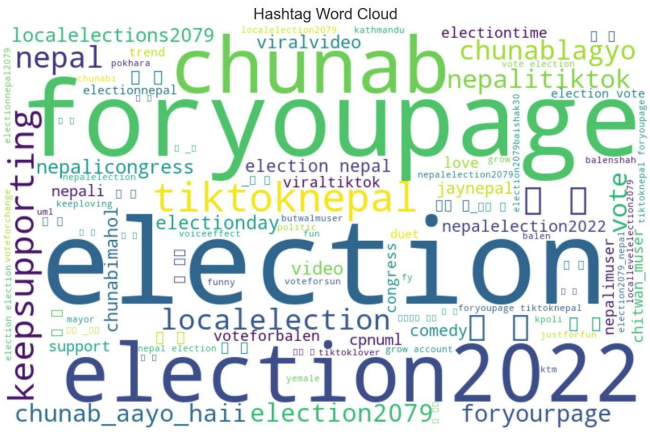

Figure 6.4: Top Filtered hashtags after removing trending terms

6.2 Virality Prediction Results

This section presents the results for our first research question, which aimed to study virality prediction using pre-upload content, platform, and sender characteristics. To focus on pre-upload features, post-upload engagement metrics (e.g., views, likes) were excluded. Results are organized into model performance, feature importance, ablation study, error analysis, and explainability using SHAP.

6.2.1 Model Performance

RQ1 was first explored through a conventional train-test model comparison and was then re-estimated using a leakage-safe, repeated cross-validation pipeline reported in the discussion and conclusion. The single-split comparison below should therefore be read as a baseline benchmark rather than the final inferential basis of the chapter. Random Forest, XGBoost, and LightGBM were trained with default parameters on a stratified train-test split (70% train, 30% test, n=890 test samples). Performance was evaluated using accuracy, macro average F1-score, and AUC-ROC (one-vs-rest).

Performance metrics included accuracy, macro F1, and class-specific F1 scores for three virality classes: low (0), mid (1), and high (2). The baseline accuracy for random guessing in a three-class problem is 33.33%.

The comparison (Table 6.1) showed that LightGBM achieved the highest accuracy (0.61) and AUC-ROC (0.7972), followed closely by XGBoost (accuracy: 0.60, AUC-ROC: 0.7948), while Random Forest lagged (accuracy: 0.58, AUC-ROC: 0.7526). All models struggled with medium virality (Class 1, F1-scores: 0.48–0.52), but performed better on the low- and high-virality classes. XGBoost was retained for interpretability and comparison with the original workflow, but the main substantive conclusion of RQ1 rests on the later refactored pipeline: sender variables dominate predictive performance across target definitions, while content-only and platform-only models are substantially weaker.

Table 6.1: Model Comparison Results (Default Parameters)

Model	**Accuracy**	**Macro Avg F1-score**	**AUC-ROC (OvR)**
Random Forest	0.58	0.58	0.7526
XGBoost	0.60	0.60	0.7948
LightGBM	0.61	0.60	0.7972

6.2.2 Performance of the Tuned XGBoost Model

The tuned XGBoost model was evaluated on the same test set (n=890), achieving an accuracy of 0.60, a macro average F1-score of 0.60, and an AUC-ROC of 0.7871. This remains useful as a descriptive benchmark and for SHAP-based interpretation, but it should be read alongside the refactored cross-validated results rather than as the sole performance estimate for RQ1. Detailed performance metrics are presented in Table 6.2.

Table 6.2: Classification Report for Tuned XGBoost Model

Class	**Precision**	**Recall**	**F1-score**	**Support**
0 (Low-Viral)	0.56	0.72	0.63	294
1 (Mid-Viral)	0.53	0.49	0.51	293
2 (High-Viral)	0.75	0.60	0.67	303
Macro Avg	0.61	0.60	0.60	890
Weighted Avg	0.62	0.60	0.60	890

The tuned model maintained stronger performance for low virality (Class 0, F1-score: 0.63) and high virality (Class 2, F1-score: 0.67) than for medium virality (Class 1, F1-score: 0.51), consistent with the baseline models. This supports the broader conclusion that the middle of the virality distribution is less separable than the extremes. In the refactored analysis, this basic pattern remains, even though the exact headline metrics vary by target definition and validation scheme.

6.3 Feature Importance Analysis

SHAP (SHapley Additive exPlanations) values were used to assess feature importance in the tuned XGBoost benchmark model, measuring the average impact of each feature on the model's predictions across the three classes on the full test set (n=890). These plots are included as illustrative model diagnostics; the main RQ1 conclusion about feature-group importance rests on the refactored cross-validated comparisons reported elsewhere in the thesis. The top 20 features are shown in the SHAP summary plot (Figure 6.5), and mean SHAP values for feature groups are presented in Table 6.3

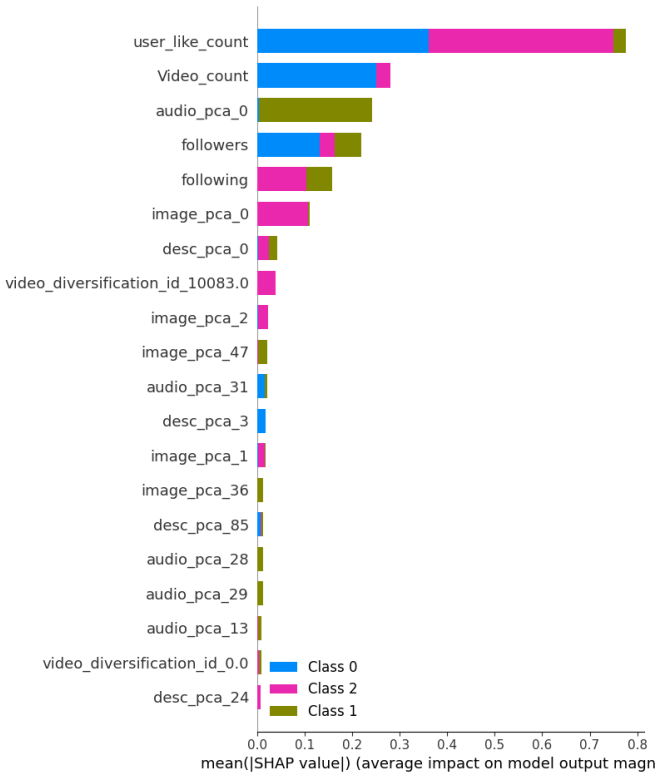

Figure 6.5: SHAP summary plot of top 20 features impacting virality prediction, with contributions to low (Class 0, blue), medium (Class 1, green), and high (Class 2, pink) virality.

6.3.1 Sender Characteristics (Mean SHAP: 0.0480)

Sender features were the dominant predictors, with a mean SHAP value of 0.0480.

- `user_like_count` (mean |SHAP| $\approx$ 0.80): Creators' total likes were among the strongest contributors to high-virality predictions (Class 2), consistent

Table 6.3: Mean SHAP Values for Feature Groups

Feature Group	Mean SHAP Value
Audio	0.0013
Image	0.0010
Text	0.0007
Sender	0.0480
Platform	0.0004

with the broader pattern that established creators had an advantage in this dataset.

- `video_count` (mean |SHAP| $\approx$ 0.40): Prolific creators' posting histories were associated with higher predicted virality across classes.
- `followers` (mean |SHAP| $\approx$ 0.3): Follower count contributed more to high-virality predictions (Class 2) than to low-virality predictions (Class 0).

6.3.2 Content Characteristics

Content embeddings dominated the top 20 features, though with smaller individual impacts than sender features:

- `audio_pca_0` (mean |SHAP| $\approx$ 0.35): The first audio PCA component notably impacted medium virality (Class 1), suggesting semantic audio content (e.g., election messages) drove moderate engagement.
- `image_pca_0`, `image_pca_1`, `image_pca_2` (mean |SHAP| $\approx$ 0.05–0.2): Multiple image PCA components appeared, showing high influence, primarily image_pca_0 for Class 2 (high-viral).
- `desc_pca_0` (mean |SHAP| $\approx$ 0.1), `desc_pca_3` (mean |SHAP| $\approx$ 0.05): Text PCA components from video descriptions affected low and high virality (Classes 0 and 2), highlighting captions' role.

Mean SHAP values were low (audio: 0.0013, image: 0.0010, text: 0.0007) due to PCA reduction (100 components per modality). Sentiment scores (`clean_audio_text_sent_*`) had minimal impact, likely distributed across embeddings.

6.3.3 Platform Characteristics (Mean SHAP: 0.0004)

Platform features had the smallest impact:

- `video_diversification_id_10083.0` (mean |SHAP| ≈ 0.1): The presence of this diversification ID in the top 10 suggests its moderate association with high virality (Class 2). Although its label was missing from the API (see Table 9.12 for details on diversification IDs and their labels), manual inspection of 30 random videos revealed themes often tied to critical speeches, public debates, and civic awareness during the election, with occasional humor and political satire, aligning with the communication styles in Table 6.4. The missing label reflects data quality issues noted in the Limitations 8.1, such as incomplete video metadata.

- `video_diversification_id_0.0` (mean |SHAP| ≈ 0.05): The default category (2,274 instances) had a minor effect, mainly for low virality (Class 0).

The limited impact of platform features may partly reflect the narrow and noisy platform metadata available for this study, including sparse diversification labels and incomplete API fields, as discussed in the Limitations section.

Table 6.4: Top Communication Styles for `video_diversification_id = 10083.0`

Style	**Count**
Critique/Frustration	103
Event Coverage	59
Civic Awareness	37
Comedic	28
Charisma/Leadership	19
Musical/Entertainment	14
Interactive	7
Communal	6
Personal Vlog/Family	4

6.3.4 Local Explanation of a High-Virality Prediction

A local explanation using a SHAP force plot was conducted for a test sample (index 88, $n = 890$, video ID: 7093702566613110043) correctly predicted as high-viral (Class 2) by the tuned XGBoost model. The plot (Figure 6.6) shows feature contributions to the Class 2 prediction, starting from a base log-odds of 0.50 and reaching a final prediction value of 0.71.

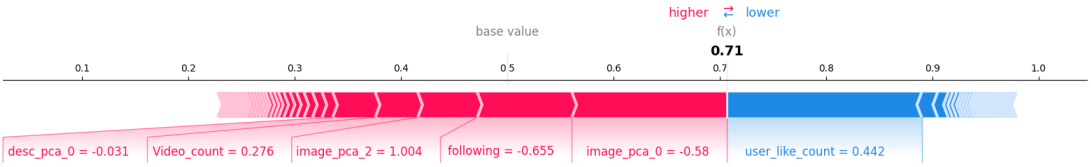

Figure 6.6: SHAP force plot for video ID 7093702566613110043, showing feature contributions to high-virality (Class 2) prediction.

Positive Contributions (Toward Class 2):

- `Video_count` = 0.276 (SHAP: +0.276): The creator's high video count strongly favored high virality (mean |SHAP| ≈ 0.5).
- `image_pca_2` = 1.004 (SHAP: +1.004): The second image PCA component, tied to visual elements (e.g., news broadcast with airplane inset), boosted the prediction.
- `user_like_count` = 0.442 (SHAP: +0.442): The creator's total likes reinforced high virality (mean |SHAP| ≈ 0.8).

Negative Contributions (Against Class 2):

- `desc_pca_0` = -0.031 (SHAP: -0.031): The video description's first PCA component (e.g., hashtags like #Localelections2079) slightly reduced virality likelihood.
- `following` = -0.655 (SHAP: -0.655): The creator's smaller following count limited virality (mean |SHAP| ≈ 0.3).
- `image_pca_0` = -0.58 (SHAP: -0.58): The first image PCA component opposed high virality.

(see Appendix 9 for more detail about this sample)

6.4 Error Analysis

An error analysis was conducted on a stratified test sample ($n = 890$) to identify misclassification patterns in the tuned XGBoost benchmark model (accuracy: 0.60, AUC-ROC: 0.7871) for ternary virality classification (0 = low, 1 = medium, 2 = high). Three specific cases and a general pattern of class boundary errors were examined to understand the model's behavior.

Case 1: Overprediction of Medium Virality as High A medium-viral video (ID: 7095593756967046426, true label: 1) with 2,501 views, 232 likes, 3,161 followers, a creator `user_like_count` of 19,854, and `Video_count` of 155 was mispredicted as high-viral (Class 2). The model's reliance on `user_like_count` (mean |SHAP| ≈ 0.8) and `Video_count` (mean |SHAP| ≈ 0.5), which strongly influence high-virality predictions, likely drove this overprediction, as the creator's high engagement history overshadowed the video's moderate engagement metrics.

Case 2: Underprediction of High Virality as Medium A high-viral video (ID: 7089029999780498715, true label: 2) with 2,820,140 views, 161,239 likes, 2,382 comments, 2,423 shares, 49,400 followers, a `user_like_count` of 1,300,000, and `Video_count` of 1,876 was mispredicted as medium-viral (Class 1). Despite exceptionally high engagement metrics, the prediction may have been moderated by content features favoring medium virality (e.g., `audio_pca_0`, mean |SHAP| ≈ 0.3). Although `user_like_count` and `Video_count` strongly support high-virality predictions, the model may struggle to capture the combined effect of extreme engagement values, possibly due to the limited impact of content embeddings (mean SHAP: audio 0.0013, image 0.0010).

Case 3: Misclassifications Between Adjacent Classes Misclassifications were most frequent between adjacent classes, particularly low (Class 0) and medium (Class 1) virality, with 62% of Class 1 errors misclassified as Class 0 and 48% of Class 0 errors as Class 1, highlighting the model's challenge in distinguishing nuanced engagement levels, as evidenced by the low F1-score for Class 1 (0.51).

6.5 Communication Styles and Political Content Themes

To examine how communication styles and political content themes relate to virality on TikTok during Nepal's local elections, we first estimate Ordinary Least Squares (OLS) models as a descriptive baseline and then conduct explicit causal feasibility diagnostics—including within-sender assessment and propensity score analysis—to evaluate whether causal interpretation is supported by the data. The main RQ2 specification is a reduced additive OLS model with communication-style indicators, political content-theme indicators, and controls for log-transformed follower count, author verification status, and days before election day. A fully interacted style–theme model is retained only as a sensitivity analysis because many interaction cells are sparse and the resulting design matrix is unstable. Reduced-model coefficients are reported in Appendix 9.10; the full interaction specification is reported in Appendix 9.9 and Appendix 9.11.

6.5.1 Model Performance

The reduced additive model explained approximately 19.5% of the variance in virality ($R^2 = 0.195$, Adjusted $R^2 = 0.187$), a reasonable fit given the high behavioral variability in social media data. The model was statistically significant ($F = 24.39$, $p < 5.29e-61$) and numerically more stable than the fully interacted alternative. After excluding four videos with invalid style code values, the reduced model uses 1,526 observations and has a condition number of approximately 206. The fully interacted sensitivity model can also be estimated on the cleaned data, but it has a notably higher condition number (approximately 650), higher variance inflation for several terms, and heavier reliance on sparse cells. For this reason, the additive specification is treated as the main descriptive model for RQ2.

6.5.2 Dataset and Category Inclusion

The filtered dataset from the sample comprised 1,530 TikTok videos, with political content identified through keyword and theme-based labeling. Five political content categories were included in the analysis, also their name along with their election symbol in Nepali script:

- Independent candidates (e.g., Balen Shah लौरो/स्वतन्त्र) — count: 232

- CPN (UML) (सूर्य/एमाले) — count: 271
- Nepali Congress (रुख/काँग्रेस) — count: 256
- CPN (Maoist) (हँसिया हतौडा/माओवादी) — count: 84
- General election-themed content (non-party affiliated) — count: 687

Of the 2,964 sampled videos, 1,530 were retained after restricting the analysis to the five political content categories. First, videos assigned to non-political content themes via zero-shot classification—including Bhojpuri/Maithili songs ($n \approx 410$), humorous or satirical content not tied to specific parties ($n \approx 320$), patriotic songs ($n \approx 85$), and unclassified speech or explanatory content ($n \approx 619$)—were excluded to maintain thematic focus on identifiable political actors and address severe class imbalance that would otherwise distort coefficient estimates. These non-political videos were retained separately for exploratory style distribution analysis to contextualize broader communication strategies. Second, videos whose content theme could not be reliably assigned through keyword and theme-based labeling were excluded. For the main additive OLS model, four additional videos with invalid style code values were dropped, leaving 1,526 observations. Certain style–theme combinations, such as Comedic–Maoist ($n = 5$), remain very sparse and motivate treating the fully interacted model only as a sensitivity analysis.

Summary statistics (Table 6.5) reveal distinct engagement patterns across content themes. Independent candidate content achieved the highest mean virality ($z_{bc_{\text{virality}}} = 0.4949$) and video comment count (89.94), reflecting strong audience interaction. Maoist content also showed elevated virality ($z_{bc_{\text{virality}}} = 0.1411$) and comments (66.20), though with fewer followers (-0.0764). Election-themed content had the lowest virality ($z_{bc_{\text{virality}}} = -0.1256$) and comments (17.85), while Congress and UML content showed negative virality scores (-0.0740 and -0.0790, respectively). Posting timing varied, with Maoist content posted furthest from election day (mean = 13.85 days) and Election-themed content closest (mean = 7.07 days).

Table 6.5: Summary Statistics by Content Theme

Content Theme	$z_{bc_{\text{virality}}}$	**Followers**	**Video Comment Count**	**Days Before Election**
Congress	-0.0740	-0.1434	22.74	9.50
Election	-0.1256	-0.1035	17.85	7.07
Independent	0.4949	0.3478	89.94	9.43
Maoist	0.1411	-0.0764	66.20	13.85
UML	-0.0790	-0.1185	29.71	10.62

6.5.3 Main Effects

The reduced additive regression (see Appendix 9.10) revealed significant main effects for content themes and control variables. Coefficient tests are reported with HC3 robust standard errors:

- **Independent candidate content** was positively associated with virality compared to the baseline election-themed content ($\beta = 0.3046$, $p < 0.001$), reflecting strong engagement even after accounting for follower reach and posting timing.
- **Communication styles**: Relative to the reference category of *Civic Awareness*, *Charisma/Leadership* ($\beta = 0.3867$, $p = 0.003$) and *Critique/Frustration* ($\beta = 0.3577$, $p = 0.012$) were positively associated with virality in the additive model. These coefficients should be interpreted cautiously, however, because the within-creator and overlap diagnostics indicate that style is highly endogenous to specific actors and contexts.
- **Control variables**: Larger follower counts ($\beta_{\text{log(followers)}} = 1.0313$, $p < 0.001$) and posting closer to election day ($\beta = 0.0058$, $p = 0.016$) were positively associated with virality. The coefficient on verified creators was negative ($\beta = -0.8951$, $p = 0.005$), but this term should be interpreted cautiously because verified accounts are very rare in the dataset.

6.5.4 Interaction Effects

Interaction coefficients are reported only as a sensitivity analysis in the appendix. Even after cleaning the coding and fixing the reference categories, the fully interacted style–theme model remains less stable than the additive specification because several combinations are sparse and some terms exhibit elevated collinearity. Accordingly, no interaction coefficient is treated as a core RQ2 finding. The more reliable conclusion is that main effects—especially independent-candidate content and follower reach, and more cautiously additive style differences relative to Civic Awareness—account for more of the observed variation than any single style–theme interaction.

Style–Theme Distribution

Exploratory analysis of style–theme distributions (Fig. 6.8) revealed strategic content framing:

- **Independent candidate content** frequently employed *Charisma/Leadership* (28.4%), often integrating rap or hip-hop to blend performance with political branding.
- **Traditional party content** (UML, Congress, Maoist) predominantly used *Music/Entertainment* (42–48%) and *Event Coverage*, featuring campaign songs, rallies, and symbolic imagery (e.g., Tree/रुख for Congress).
- **Funny content** was primarily *Comedic or Political Satire* (63.7%), while *Patriotic* and *Speech* content leaned toward *Civic Awareness* and *Critique/Frustration*.
- **Non-political content** paired with folk, instrumental, or romantic Nepali pop music, emphasizing cultural appeal.

Note: The dotted red line in Fig. 6.8 separates Political Party Support/Campaign Songs from other audio/content themes.

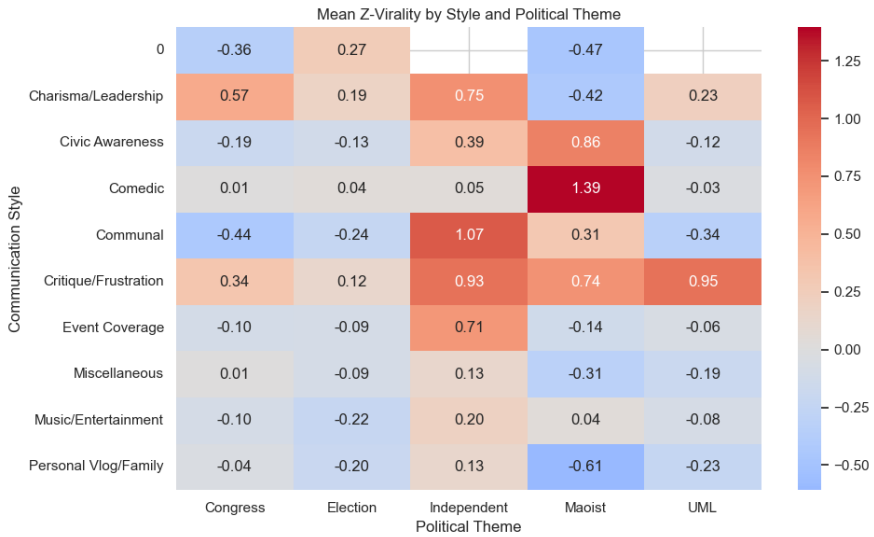

Figure 6.7: Mean z-transformed virality scores by communication style and political content theme.

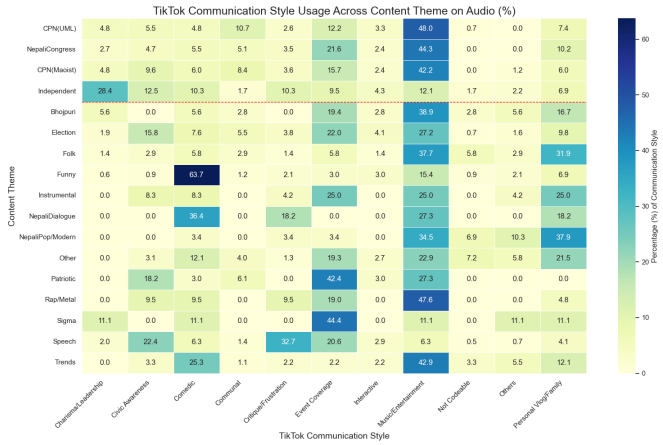

Figure 6.8: Heatmap of communication style and content theme distributions.

6.6 Causal Feasibility Diagnostics (Within-Sender and Overlap Checks)

RQ2 asks how communication styles and political content themes are associated with virality. Addressing this question causally would require comparing videos that are similar in sender characteristics and context but differ in communication style. The following diagnostic analyses evaluate whether such counterfactual comparisons are empirically plausible in the observed data.

Table 6.6: Propensity Score Distribution for Charisma/Leadership

Sample	N	Mean	Median	25%	75%	Min	Max
Full sample	1,526	0.067	0.00002	∼0	0.0036	∼0	1.00
Trimmed sample	193	0.190	0.081	0.040	0.212	0.020	0.789

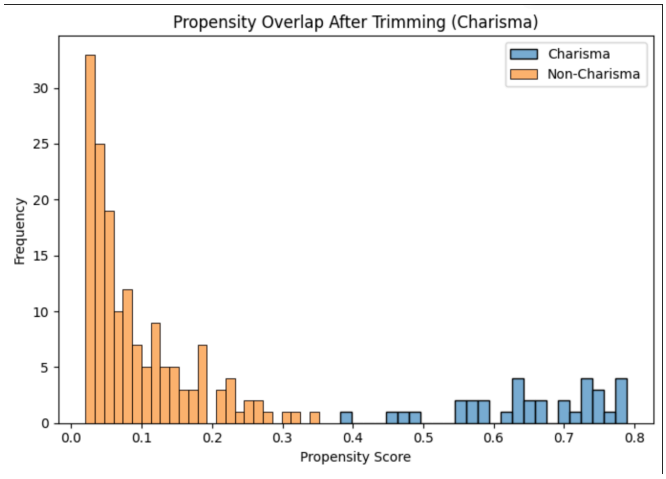

Figure 6.9: Propensity score overlap for Charisma/Leadership after trimming.

The figure plots estimated probabilities of adopting a Charisma/Leadership style based on pre-treatment sender and timing covariates. Treated and untreated distributions exhibit minimal overlap, indicating a violation of the common support assumption and limiting causal identification.

Propensity score diagnostics (Table 6.6; Figure 6.9) reveal severe imbalance in style assignment probabilities, with minimal overlap between Charisma/Leadership and non-Charisma videos even after trimming. Balance statistics (Table 6.7) further indicate that treated and control groups differ systematically along key sender characteristics. These results violate the common

Table 6.7: Covariate Balance After Trimming

	Means		**Medians**	
Covariate	Non-Charisma	Charisma	Non-Charisma	Charisma
Days before election	9.55	9.82	6	6
Verified	0.063	0.000	–	–
Followers (std.)	0.144	-0.057	-0.247	-0.247
Account age (std.)	0.113	0.079	0.021	0.263

support assumption required for causal identification, motivating an associational interpretation of the OLS estimates. *Interpretation note:* Although this section is motivated by causal language in RQ2, diagnostic analysis demonstrates that communication styles are highly selective and lack sufficient counterfactual overlap. Accordingly, all results below are interpreted as conditional associations rather than causal effects.

Importantly, the absence of causal identifiability is itself substantively informative. Diagnostic analysis indicates that communication styles such as Charisma/Leadership are not interchangeable tactics but are structurally tied to specific political actors and roles. This suggests that style operates less as a manipulable treatment and more as a signal of political positioning within the TikTok attention economy.

7. Discussion

This study asks two simple questions about Nepal's 2022 local elections on TikTok: can we predict which political videos will attract attention, and are some communication styles or political themes linked to higher engagement than others? The answer is mixed. Some patterns are clear, but the data also place real limits on what can be claimed.

The central empirical pattern is straightforward. Virality is shaped more by *who* is posting than by content features alone. In other words, political attention on TikTok in Nepal is not only about message design. It is also about pre-existing visibility, audience size, and who is already well positioned to be seen.

7.1 Predictability of Political Virality (RQ1)

The refactored RQ1 pipeline shows that political virality is moderately predictable from pre-upload features, but that prediction is driven primarily by sender characteristics. Under the primary, less view-dependent target, the best full model achieves an AUC-ROC of approximately 0.80, while a sender-only model performs nearly as well. By contrast, content-only and platform-only models perform substantially worse. This pattern is more important than the exact headline metric: it indicates that baseline creator capacity carries far more predictive information than multimodal content features on their own.

For a non-technical reader, the meaning of this result is simple: the model can tell something useful about future visibility, but it mostly does so by learning which accounts were already strong to begin with. This also helps explain why medium virality remains difficult to classify. Videos in the middle of the distribution do not look as clearly different from one another as the lowest- and highest-performing videos.

7.1.1 Sender Capacity

Sender characteristics, particularly prior creator engagement, posting history, and follower count, dominate the predictive models. Substantively, this suggests that political virality on TikTok in Nepal is not well explained by content features alone. Videos are embedded in pre-existing creator trajectories, so account-level visibility and accumulated audience attention matter before a specific video is even evaluated.

At the same time, this result should not be over-read as proof of TikTok's hidden algorithm. What the data show is simpler and safer: observed political visibility is unequally distributed. That inequality may reflect some combination of platform recommendation, existing follower base, cross-platform promotion, and offline public prominence.

7.1.2 Limits of Predictive Modeling

The model is useful, but it is not magic. It captures some broad patterns well, especially around creator scale, but it is weaker at capturing context-specific meaning. That matters in Nepal, where language mixing, regional dialects, humor, and local political symbolism are often central to how content is understood. Frequent confusion between low and medium virality reflects this limitation: not every politically important difference is easy to represent as a machine-readable feature.

7.2 Revisiting RQ2: What Can (and Cannot) Be Claimed

This study also examined how communication styles and political content themes are associated with virality on TikTok. Here the key point is caution: the analysis does not support a causal interpretation of style coefficients.

Diagnostic analyses—including within-sender assessment and propensity score–based overlap checks—show that communication styles such as *Charisma/Leadership* or *Comedic* content are highly selective and lack sufficient counterfactual overlap. Consequently:

- These styles cannot be treated as manipulable treatments.
- Independent causal effects on engagement cannot be reliably estimated.
- Results are interpreted as **conditional associations** describing the observed political communication ecosystem.

7.3 Communication Style as a Structural Signal

A central insight is that communication style is difficult to interpret as an independent driver of virality. In the reduced additive model, *Charisma/Leadership* and *Critique/Frustration* are positively associated with virality relative to *Civic Awareness*. However, these associations become harder to sustain as standalone substantive claims once richer interaction structures, overlap diagnostics, and within-creator sensitivity checks are considered. The within-creator analysis in particular shows little evidence that switching style within the same creator is associated with large systematic changes in virality once creator-level differences are absorbed.

In simpler terms, style seems to travel with the speaker. It is not just a tool that any political actor can pick up and use with the same effect. For example, *Charisma/Leadership* is disproportionately associated with independent candidates (primarily associated with Balen Shah) and nearly absent among legacy-party accounts. Its apparent effectiveness depends on:

1. **Political legitimacy:** Who is allowed to speak in a certain way.

2. **Prior visibility:** The baseline reach of the creator.

3. **Audience trust:** The perceived authenticity of the performance.

7.4 Independent Candidates: Virality and Authenticity

Content related to independent candidate (mainly Balen Shah) followed by Harka Sampang, and Gopi Hamal exhibits substantially higher engagement than legacy-party content. Independent candidate videos achieve the highest mean virality scores, even when controlling for follower counts.

Charismatic performance—expressed through rap, hip-hop, and symbols like the *lauro*—is a defining characteristic of prominent independent-candidate content in the dataset. The evidence is strongest for the descriptive claim that this style clusters around independent actors and accompanies higher engagement in that subset, not for the stronger claim that charisma itself independently raises virality across actors.

The more defensible interpretation is therefore relational rather than tactical. Independent-candidate content appears to combine a favorable political moment, recognizable symbols, and creator-specific authenticity in ways that align with higher engagement. In a low-income but highly connected country like Nepal, this matters because social media can help new political actors bypass parts of the traditional media system. But the findings also suggest that this opening is not equal for everyone; visibility still depends heavily on who already has reach and recognition.

7.5 Legacy Parties and Conditional Resonance

Legacy parties (UML, Congress, Maoists) generally show lower average engagement. However, the interaction between *Critique/Frustration* and UML-related content is positively associated with virality.

Table 7.1: Style and Actor Correlations

Actor Group	Dominant Styles	Engagement Driver
Independent Candidates	Charisma, Performance	Authenticity & Youth Appeal
Legacy Parties	Music, Event Coverage	Critical/Confrontational Framing
General Election Content	Civic Awareness	Informational Utility

This interaction should be treated cautiously. It is substantively plausible and consistent with Affective Intelligence Theory, but it is not the basis for a broad strategic recommendation. Across the RQ2 analysis, the more robust conclusion is that theme-level differences are clearer than style-level effects, and that actor-bound content categories matter more than any single communication style coefficient.

7.6 Integrating RQ2 with RQ1

While the XGBoost model in RQ1 shows that virality is predictable, it also reveals that **sender-related features** (follower count, prior engagement) dominate performance. RQ2 clarifies this:

> RQ1 establishes that virality is predictable largely due to *who* is speaking, while RQ2 shows that *how* they speak is constrained by *who* they are.

7.7 Broader Implications

The findings carry implications across four audiences: researchers, policymakers and election authorities, citizens and civil society, and political practitioners.

For researchers, this study demonstrates that observational social media data cannot sustain strong causal claims about communication strategy effects without explicit identification testing. The propensity score diagnostics presented here—rarely conducted in political TikTok research—should become standard practice before attributing engagement differences to content choices. The open-source multimodal pipeline developed here is directly transferable to comparable Global South contexts where such analytical infrastructure is absent.

For policymakers and election authorities, the dominance of sender features

in predicting virality suggests that political visibility on TikTok is structured by unequal creator capacity, not just by the content of individual videos. In Nepal, this matters because short-video platforms can give citizens low-cost access to campaign information, political commentary, satire, and candidates who may receive less attention in legacy media. That is a real democratic opportunity. But it also creates an uneven visibility landscape in which candidates with pre-existing digital audiences may compound their reach, whether through platform recommendation, cross-platform spillover, or offline prominence. Nepal banned TikTok in 2023 and reinstated it in 2024, with debates driven largely by misinformation concerns (Lamichhane, 2024; Reuters, 2024). The present findings cannot identify TikTok's internal ranking rules directly, but they still point to clear practical priorities for Nepal.

First, the **Election Commission of Nepal** should ask platforms for more transparency during election periods. In plain terms, this means not only asking whether harmful content is being removed, but also asking which political accounts, influencers, and campaign-related videos are receiving especially high visibility.

Second, Nepal's election authorities, public-interest researchers, and the government bodies responsible for digital policy should push for **better archival and research access**. One of the biggest limits of this thesis was that election-period platform data were incomplete and hard to verify. Without a better archive of public political content and metadata, independent scrutiny after an election remains weak.

Third, campaign oversight in Nepal should treat **digital reach as a campaign resource**. If some actors enter the campaign with much larger online audiences, then visibility is not evenly distributed from the start. This suggests a need for closer scrutiny of high-reach political amplification, including influencer promotion, coordinated visibility campaigns, and the role of large intermediary accounts.

Finally, the findings support **public media literacy efforts** in Nepal that help citizens read online popularity more carefully. A highly viewed or highly shared political video should not automatically be taken as a signal of broad public support; it may also reflect the advantage of already prominent accounts.

For citizens and civil society, the core message is plain: social media can be useful during an election because it helps people see speeches, campaign clips,

criticism, humor, and local political discussion quickly and at low cost. This is especially important in a country like Nepal, where digital platforms may be easier to access than many traditional sources of political information. But what becomes visible online is not necessarily what most people support. It is partly shaped by who already has digital reach. For journalists, watchdog groups, and ordinary voters, this means viral political content should be read with one more question in mind: who is being amplified, and who may be left out of view?

For political practitioners, the main practical implication is negative rather than prescriptive: the analysis does not justify strong claims about "what works" stylistically on TikTok. The evidence is more consistent with the view that communication styles operate as structural signals tied to specific actors and audiences. Legacy parties therefore cannot assume that copying the visible form of independent-candidate content will reproduce its engagement outcomes.

8. Conclusion

This study examined TikTok's role in shaping political attention during Nepal's 2022 Local Elections through two complementary research questions: predicting political virality using pre-upload features (RQ1), and examining how communication styles and political content themes are associated with engagement outcomes (RQ2). Situated in a low-income, multilingual, but highly digital context, the study contributes to computational political communication research while also speaking to a wider public debate about what social media means for elections in Nepal.

On RQ1, the refactored prediction pipeline shows that political virality on TikTok can be predicted with moderate success from pre-upload features, but that sender-related characteristics—prior engagement history and follower count—consistently matter more than content or platform features alone. The key conclusion is therefore about structural visibility: political attention on TikTok is not well explained as a purely content-driven process, because creator capacity carries substantial predictive information.

On RQ2, regression analysis revealed that content associated with independent candidates achieved substantially higher engagement than either general election-themed or legacy-party content—the highest mean virality score in the dataset ($z_{bc_{\text{virality}}} = 0.4949$), even after controlling for follower reach. By contrast, communication-style coefficients are unstable across model specifications and diagnostic checks. The safest interpretation is therefore that styles such as *Charisma/Leadership* are concentrated in independent-candidate content and function as actor-bound signals rather than as independently validated drivers of engagement.

Policy relevance. These findings speak directly to governance challenges that Nepal faced when it banned TikTok in 2023 and reversed course in 2024 (Lamichhane, 2024; Reuters, 2024). Short-video platforms can widen access to political information and help newer actors reach voters, which is especially important in a low-income but highly connected country such as Nepal. But content modera-

tion and platform bans address only one dimension of the problem. The deeper challenge suggested by this thesis is unequal political visibility on short-video platforms, which may reflect the interaction of creator scale, engagement dynamics, and platform design. The main policy lesson is therefore institutional rather than stylistic. For Nepal, this means that the Election Commission, digital-policy authorities, and public-interest researchers need better transparency about how political content is distributed, better archival access to election-period platform data, and campaign oversight rules that recognize digital reach as a form of political advantage. In simpler terms: the problem is not only bad content, but also unequal visibility. For citizens, the practical meaning is equally simple: a viral political message may be influential, but it is not automatically representative.

Methodological contribution. By explicitly testing and rejecting causal identification assumptions rather than silently reporting OLS coefficients as effects, this analysis provides a replicable template for political communication research in contexts where experimental or quasi-experimental designs are unavailable. The open-source multimodal pipeline, composite virality index, and communication style codebook are transferable to comparable settings. Future work should pursue longitudinal designs that track the same creators across election cycles, cross-platform comparisons that isolate TikTok-specific effects from broader social media dynamics, and—as API access matures—network-level analyses of repost and follower graph structures that the current TikTok Research API does not yet support (TikTok, 2025).

8.1 Limitations

While the baseline single-split benchmark achieved about 60% accuracy and the refactored pipeline showed moderate predictive performance across validation settings, several limitations warrant acknowledgment:

- **Predictor Variables and Sampling**: Time constraints limited processing of the full 28,000-video dataset. Stratified sampling ensured coverage across the engagement distribution, but the subset's representativeness relative to the full corpus was not fully validated. Because the sample was initially drawn using view-based strata, some degree of selection toward already visible content cannot be ruled out.
- **Audio Transcripts**:
  - Open AI's Whisper model struggled with audio mixed with sound effects, retaining only clear segments due to time limitations.
  - Manual labeling of Nepali text for contextual meanings (e.g., "party song," "humor") was applied, but transcripts in other languages (e.g., Urdu, Chinese), slang, or noisy audio posed challenges. Contextual labels (e.g., "Newari Language," "Bhojpuri/Maithili song") lacked validation.
  - Nepal's linguistic diversity led to language estimation based on the author's knowledge, reducing transcript accuracy.
  - Differing audio-video tones (e.g., fun audio vs. satirical video) confused RQ2 labeling.
  - A Cohen's Kappa score of 0.462 indicates moderate agreement for transcripts (DATAtab, 2025; Kolena, 2025), reflecting challenges for LLMs in processing low-resource languages like Nepali's dialects and noisy social media content (ICUC Social, 2025; Lean Mean Learning Machine, 2023).
- **Video Descriptions**:
  - Semantic interpretation of Nepali humor, dialects, and mixed Romanized English-Nepali text was challenging. Names like सागर (Sagar, mistranslated as "Ocean") or बसन्ते (Basante, "Spring") in campaign songs required manual correction to capture satirical intent.

- Accuracy was assessed on a 5% sample, potentially unrepresentative of the full dataset.
- Empty descriptions and Lang Detect API errors misidentifying mixed text as English skipped translations, affecting embedding accuracy.

- **Image Analysis**:
  - Image analysis missed contextual nuances (e.g., running mistaken for walking in screenshots).
  - Using 3-second clips, inspired by TikTok marketing, lacked academic validation in this context.
- **Style_1_Label**:
  - The custom codebook struggled with blended social media content (e.g., humor, music, vlogging), limiting nuanced meaning capture.
  - Annotator bias led to subjective categorizations (e.g., rally videos as "communal" [Category 4], "event coverage" [Category 9], or "personal vlogging" [Category 3]). High Cohen's Kappa (> 0.60) was sample-based and not standardized for Nepali election contexts.
  - Communal and event coverage labels were often conflated, as communal themes appeared in documentary-style or rally footage.
  - Discrepancies between the author's codebook and annotators were resolved only for significant differences, risking minor inconsistencies.
- **Content_Theme**:
  - Zero-shot labeling struggled with humorous multi-party support videos, forcing single-theme selection.
  - Symbolic imagery (e.g., money in a child's mouth) was insufficient for clear thematic categorization.
- **Within-Creator Variation for RQ2**:
  - Although a subset of creators posted videos in multiple styles, within-creator variation remained limited relative to the full political sample. This constrains the extent to which communication styles can be compared net of stable creator characteristics.
  - As a result, RQ2 is better suited to describing actor-linked style patterns than to isolating the independent contribution of style alone.
- **Platform Features**:

- The `video_diversification_id` (e.g., `id_10083.0`) had sparse, missing API labels, complicating content type interpretation (Table 9.12). Overfitting to dominant IDs (e.g., `video_diversification_id_0.0`) may reduce generalizability for impactful, less frequent IDs influencing high virality (Class 2).

- **Transformer Models for Low-Resource Settings**:
  - Open-source models like Wiseyak (Duwal et al., 2024) and Nepal-iBERT (Pudasaini et al., 2023) were limited to Devanagari text, and still surprisingly underperformed on Devanagiri setting, let alone mixed Romanized Nepali-English content. SeamlessM4T (Elbayad et al., 2023; Facebook Research, 2023; Hugging Face, 2024) excelled in transliteration and translation than Nepalese fine-tuned counterpart (for Devanagiri). Comparative analysis among these models were not acknowledged due to time constraints.
  - No Hugging Face model fully addressed transliteration, translation, and semantic embedding for code-switched Nepali-English text. SeamlessM4T-medium handled Devanagari better, but struggled with code-switching, while OpenAI's API proved more accurate (Reddit, 2023).

- **Data Limitations**:

  The TikTok Research API, while enabling access to public data, suffers from archiving delays and incomplete historical records, limiting accurate view and engagement counts for Nepal's 2022 election videos (TikTok, 2025). As a relatively new API, it offers room for improvement in data reliability and historical access. Additionally, unofficial APIs (Bukold, 2025), reliant on web-inspect elements, may have some inaccuracies, as manual inspections revealed discrepancies in variables like `video_is_ai_gc`, where it is null in metadata, but exists in the actual video.

8.2 Future Research

Future research could address these limitations by:

- Scaling up to the full 28,165-video dataset would enable finer-grained virality levels and more nuanced category distinctions.
- Developing time-series models to capture evolving virality dynamics across campaign cycles.
- Comparing virality patterns across platforms (e.g., Facebook(Meta) platform, YouTube Shorts, Instagram Reels) to assess platform-specific effects. Facebook still continues to be the most dominant social media platform in Nepal, used by half of country's population (NapoleonCat, 2025).
- Enhancing comment analysis, including sentiment, to better understand virality impact despite time constraints on initial collection.
- Employing fine-tuned multilingual embeddings for Nepali and Romanized text to enhance semantic capture.
- Using network analysis to explore virality spread, requiring follower graphs and repost chains. While the current TikTok API provides follower and repost counts, it does not include lists of followers or reposters (likely due to privacy restrictions, despite followers being manually viewable), necessitating future API enhancements to enable such analysis.
- Exploring hybrid models with attention mechanisms to improve medium virality prediction.

By addressing these gaps, researchers can further elucidate the role of popular social media platforms such as TikTok in shaping political communication, particularly in diverse, multilingual contexts like Nepal.

9. Appendix

Dataset Summary

Metric	Count
Total Videos	28,165
Total Transcripts	28,165
Total Comments	307,156
Total Views	186,348,152
Total Shares	214,808
Total Likes	14,334,935
Total Saves/Collects	83,422
Unique Users	19,993
Verified Users	9

Table 9.1: Dataset summary statistics

Keywords and Hashtags for Nepal Local Election 2022

- election
- elections
- locallevelelection2079
- स्थानी_चुनाव_२०७९
- चुनाव
- localelection2079
- मतदानगरौं
- NepalElection
- election2079
- election2022
- nepalelection2022
- localelections2079
- electionday
- election2079_nepal
- election2079baishak30
- elections2022
- nepalvotes2022
- local_election
- chunab
- chunablagyo

A.1 Audio PCA Samples (`audio_pca_0` Extremes)

Table 9.2: Top Values of `audio_pca_0` and Corresponding Content

audio_pca_0	Transcript (Nepali)	Virality Label
0.957	*Nepali Instrumental/Flute Song*	1
0.925	*Nepali UML song*	0
0.917	*Bollywood Instrumental Song*	0
0.911	*Nepali trend song*	1
0.906	*Bollywood Remix Song*	1

Table 9.3: Bottom Values of `audio_pca_0` and Corresponding Content

audio_pca_0	Transcript (Nepali)	Virality Label
-0.314	अब पक्कै बन्दैछ, माया कांग्रेस	1
-0.314	यदि एक अध्यक्षले तीन लाख...	1
-0.314	मत्दान गरौँ जानेरा, बुजेरा...	0
-0.314	मुला छमता कियो भनेको मैले...	0
-0.314	सपुर्ण माया नमस्कार, म भोजराज...	1

A.2 Virality Distribution by Diversification ID

Table 9.4: Selected Diversification IDs and Virality Class Proportions

ID	Class 0 (%)	Class 1 (%)	Class 2 (%)
0.0	35.7	35.8	28.5
10003.0	0.0	0.0	100.0
10012.0	0.0	0.0	100.0
10014.0	11.1	0.0	88.9
10017.0	66.7	22.2	11.1
10025.0	0.0	100.0	0.0
10083.0	7.9	14.4	77.7
10088.0	0.0	0.0	100.0

A.3 Error Rates by Sender Features

Table 9.5: Error Rates by Sender Characteristics

Feature	Error Rate (%)
Author Verification	
Verified	0.0
Unverified	37.1
Follower Tier	
Low	44.8
Mid-Low	38.3
Mid-High	38.7
High	24.7

A.4 A Sample Case Study

Sample Details:

- **Virality Score**: 3.6679
- **True/Predicted Label**: 2
- **Content Description**: The video humorously comments on money found aboard a Buddha Air flight, making a political reference to Janardan Sharma during Nepal's 2022 elections. The audio includes comedic Nepali dialogue, the description uses hashtags (e.g., #Localelections2079), and visuals resemble a news broadcast with an airplane image inset.

Context and Interpretation: The video's high virality is plausibly related to its comedic tone, political relevance, and active creator profile (evidenced by metrics like `user_like_count` and `Video_count`). The benchmark model predicted its virality score correctly, consistent with the broader SHAP pattern that sender features and some image embeddings carried substantial predictive information. Description text and network features played a more limited role in this example.

A.5 Development of TikTok Communication Style Codebook

This appendix outlines the development and structure of the communication style codebook used to categorize TikTok videos for RQ2 in the analysis of political communication during Nepal's 2022 Local Elections. The codebook was iteratively refined to capture the nuanced and context-specific nature of TikTok's political discourse in Nepal, building on established frameworks and empirical observations.

Development Process The initial codebook was adapted from the communication style classification framework proposed by Umansky and Pipal (Umansky & Pipal, 2023), which analyzed political communication on TikTok and included categories such as Comedic, Documentary, Communal, Explanatory, Interactive, Meta, Other, and Not Codeable. This framework provided a theoretical foundation but required adaptation to address the unique characteristics of election-related content in Nepal.

Through an iterative annotation process involving a stratified sample of TikTok videos, the codebook was expanded to incorporate additional styles that emerged as salient in the dataset. These included Musical/Entertainment, Charisma/Leadership Portrayal, Civic Engagement/Awareness, and Populist Critique/Frustration. The refinement process aimed to balance theoretical grounding with inductive insights, ensuring the codebook captured the richness of multimodal political content while remaining practical for annotation. In cases of overlapping styles, annotators prioritized the dominant communicative intent, determined by integrating visual, auditory, and textual cues, as perceived by a typical viewer.

The final codebook, presented in Table 9.6, reflects both the adaptation of prior research and context-specific adjustments derived from direct engagement with Nepal's TikTok political landscape.

Codebook Structure Table 9.6 details the communication style codes, their labels, definitions, and illustrative clues observed in video content or transcripts. Each style is designed to encapsulate a distinct mode of political communication, facilitating the analysis of style–theme interactions in the OLS regression model (Appendix 9.9).

Table 9.6: TikTok Communication Style Codebook for Nepal's 2022 Local Elections

Code	Label	Definition	Clues in Video/Transcript
1	Comedic	Uses humor, parody, satire, or comedic characters to convey political messages, often leveraging viral trends or meme formats.	Emojis, punchy phrases, satirical portrayals of politicians, voiceovers of viral characters (e.g., Bhelu Baje), comedic sketches with social messages.
2	Musical/Entertainment	Incorporates musical elements such as songs, rap, Bollywood/South Indian dialogues, or traditional folk music to entertain or subtly promote political messages.	Background music, rhythmic speech, dancing, election campaign songs, acting over Nepali/Hindi/South Indian movie dialogues, folk songs. Focus is more on music or enjoying music or entertainment genre content.
3	Personal Vlog/Family	Depicts personal or family life with loose connections to political content, often featuring casual vlogging during election-related events.	Selfie vlogs, children playing or casually appealing for votes, family bonding with election references.

Table 9.6: TikTok Communication Style Codebook (Continued)

Code	Label	Definition	Clues in Video/Transcript
4	Communal/Community Identity	Emphasizes collective identity, shared pride, or belonging to a hometown, ethnic group, or cause, highlighting emotional or physical gatherings for political action.	"We the people" rhetoric, celebration of local traditions, community gatherings, group best-wishes, emotional bonding during rallies.
5	Charisma/Leadership Appeal	Showcases an individual leader's strength, savior-like image, or charismatic appeal, often integrated with music or group settings.	Heroic edits, repeated slogans (e.g., #VoteForBalen), energetic or motivational background music, leader surrounded by supporters, "idolization" feel.
6	Interactive/Participatory	Explicitly invites user engagement through comments, votes, duet collaborations, or reaction videos.	"What do you think?", "Comment below!", direct questions, TikTok comment replies, Duet chains, candidates replying to individual TikToks.
7	Critique/Frustration	Critiques elites, government systems, or the political status quo, appealing to ordinary people's struggles with frustration or anger.	"घरमै ३० हजार?", corruption allegations, sad or angry tones, poetry/rap with critique, blaming inefficiency, emotional storytelling of hardships, media cases (e.g., Rabi Lamichhane's exposés).

Table 9.6: TikTok Communication Style Codebook (Continued)

Code	Label	Definition	Clues in Video/Transcript
8	Civic Engagement/Awareness	Educates or appeals to voters' sense of civic responsibility, often using a serious, informative tone.	Captions like "Vote smart", educational visuals of ballot papers, voting process explanations, voter awareness campaigns, hashtags promoting participation.
9	Event Coverage	Provides direct, often raw footage of election-related events with minimal editorial input, focusing on rallies, protests, or behind-the-scenes activities.	Unedited rally footage, behind-the-scenes voting booth setups, interviews at public events, serious event documentation.
10	Other	Election-related content not fitting defined categories, often with mixed styles or secondary to non-political content.	Clothing ads combined with voting calls, casual vlogs with unclear political communication, confusing or multistyle videos.
99	Not Codeable	Content irrelevant, missing, or unusable due to poor quality or technical issues.	Private/deleted videos, foreign TV series clips, inaudible sound, irrelevant non-election posts.

A.6 Content Themes for Zero-Shot Classification

Table 9.7 presents the content themes used for zero-shot classification of TikTok videos in the analysis of political communication during Nepal's 2022 Local Elections. These themes were derived to capture the diverse range of political and non-political content, facilitating the style–theme interaction analysis in RQ2 (Appendix 9.9).

Table 9.7: Content Themes for Zero-Shot Classification

Theme	**Potential Content**
UML (सूर्य/एमाले)	Campaign songs or content supporting CPN (UML), often featuring the Sun symbol.
Congress (काँग्रेस/रुख)	Campaign songs or content supporting Nepali Congress, often featuring the Tree symbol.
Maoist(हँसिया/हतौडा)	Campaign songs or content supporting CPN (Maoist), often featuring the Sickle/Hammer.
Independent (लौरो)	Campaign songs or content supporting Independent Candidate, mainly Balen Shah, also featuring other youth/independents such as Harka Sampang
Political Speech	Speeches by candidates or party leaders, focusing on election campaigns or policies.
Bollywood	Songs or audio from Bollywood movies, often used for entertainment or lip-syncing.
Instrumental	Instrumental music, typically background tracks without vocals, used in various contexts.
Bhojpuri/Maithili	Songs or audio in Bhojpuri or Maithili languages, common in Nepal's Terai region.
Urdu	Songs or audio in Urdu, often reflecting cultural or poetic themes.
Sigma Sound/Trend	TikTok trends involving "sigma male" memes, often with specific background sounds.
Nepali Folk/Traditional Song	Traditional Nepali songs, often tied to cultural or festive themes (e.g., Dashain).
Funny	Humorous content, including comedic skits, parodies, or satire (political or not).

Table 9.7: Content Themes for Zero-Shot Classification (Continued)

Theme	**Potential Content**
Sad Song	Emotional or melancholic songs, often used in sentimental or dramatic videos.
Patriotic Song	Songs promoting national pride, often linked to Nepal's history or identity.
Rap Song	Rap or hip-hop music, often used by independent candidates for modern appeal.
TikTok Trends	Viral TikTok trends, including challenges, dances, or trending sounds.
Election Song	General election-themed songs, not tied to a specific party, promoting voting or democracy.
Other/Non-Political	Miscellaneous content, including non-political explanatory videos, vlogs, or ambiguous cases.

9.1 GitHub Repository Details

The source code for this project is available on GitHub at the following link: https://github.com/nimathing2052/TikTok_Prediction

If you are unable to access the repository, please feel free to contact the author for access.

Table 9.8: Comparison of sentence embedding model performance on sample Nepali election text pairs

Index pair	Similarity type	LaBSE	MiniLM-L12	all-MiniLM-L6-v2	BGE m3	Best model(s)	Rationale
1035;1293	Moderate (Congress rally, different visuals)	0.4617	0.4965	0.5465	0.4905	all-MiniLM-L6-v2	Highest mid-range score reflects shared party and activity
314;457	High (CPN UML song, different visuals)	0.8320	0.8383	0.8040	0.8780	BGE m3	Highest score reflects strong topic match
135;180	Very high (UML song, different visuals)	0.9232	0.9665	0.9398	0.9509	MiniLM-L12, BGE m3	Very high scores appropriate for near-identical topics
865;360	Low-moderate (Voting appeal vs. local election)	0.2961	0.4851	0.0879	0.5770	MiniLM-L12, BGE m3	Moderate scores capture shared election theme
890;882	Low-moderate (Voting integrity vs. election duty)	0.1150	0.3134	0.6181	0.3777	MiniLM-L12, BGE m3	Low-moderate scores appropriate; all-MiniLM-L6-v2 too high
1889;2226	Low (Generic election vs. specific Balen)	0.1047	0.3855	0.6954	0.5455	LaBSE	Lowest score appropriate; others too high
1122;1242	Low (General query vs. specific Balen)	0.1327	0.2775	0.1670	0.4773	LaBSE, all-MiniLM-L6-v2	Low scores appropriate; BGE m3 too high
180;1648	Low (UML song vs. general voting/petrol)	0.0854	0.1194	0.1850	0.3242	LaBSE, MiniLM-L12, all-MiniLM-L6-v2	Low scores appropriate
1687;1862	Low (UML vs. Balen campaign)	0.1238	0.5294	0.6342	0.3743	LaBSE	Lowest score appropriate for opposing subjects
888;1462	Moderate (Campaign comedy vs. election dialogue)	0.0651	0.3088	0.1797	0.4557	BGE m3, MiniLM-L12	Moderate scores capture thematic link
352;1331	Moderate (UML candidate vs. child supporter)	0.0229	0.2001	0.4907	0.3105	all-MiniLM-L6-v2, BGE m3	Moderate scores capture shared party theme

Note: Similarity types describe topical overlap and visual differences. Models include LaBSE, MiniLM-L12, all-MiniLM-L6-v2, and BGE m3. UML refers to Unified Marxist-Leninist party; Balen refers to a specificible specific candidate.

Table 9.9: Summary Statistics for Sensitivity OLS Regression with Full Style–Theme Interactions

Dep. Variable:	z_bc_virality (Virality z-score)	**R-squared:**	0.213
Model:	OLS	**Adj. R-squared:**	0.188
Method:	Least Squares	**F-statistic:**	8.527
Date:	cleaned sensitivity model	**Prob (F-statistic):**	8.31e-50
Time:	–	**Log-Likelihood:**	-1982.2
No. Observations:	1526	**AIC:**	4060.5
Df Residuals:	1478	**BIC:**	4316.3
Df Model:	47		
Covariance Type:	nonrobust		

Table 9.10: Reduced Additive OLS Model Predicting z_bc_virality

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	−0.0003	0.077	−0.004	0.997	−0.152	0.151
Main Effects (Style, Ref: Civic Awareness)						
Style: Charisma/Leadership	0.3867	0.131	2.950	0.003	0.130	0.644
Style: Comedic	0.1293	0.123	1.056	0.291	−0.111	0.369
Style: Communal	−0.0821	0.145	−0.568	0.570	−0.366	0.202
Style: Critique/Frustration	0.3577	0.142	2.513	0.012	0.079	0.637
Style: Event Coverage	0.0891	0.089	1.001	0.317	−0.086	0.264
Style: Miscellaneous	−0.0565	0.116	−0.487	0.626	−0.284	0.171
Style: Music/Entertainment	−0.0136	0.084	−0.162	0.872	−0.179	0.151
Style: Personal Vlog/Family	−0.0317	0.101	−0.315	0.753	−0.229	0.166
Main Effects (Content, Ref: Election)						
Content: Congress	0.0505	0.065	0.780	0.436	−0.077	0.178
Content: Independent	0.3046	0.078	3.909	0.000	0.152	0.458
Content: Maoist	0.1942	0.104	1.872	0.061	−0.009	0.398
Content: UML	0.0595	0.069	0.860	0.390	−0.076	0.195
Control Variables						
Author Verified	−0.8951	0.317	−2.824	0.005	−1.517	−0.273
Log(Followers)	1.0313	0.078	13.230	0.000	0.878	1.184
Days Before Election	0.0058	0.002	2.418	0.016	0.001	0.011

Notes: Reference category for Style is 'Civic Awareness'. Reference category for Content is 'Election'. HC3 robust standard errors are reported. The model includes 1526 observations after dropping four videos with invalid style code values.

Table 9.11: Sensitivity Interaction Effects for OLS Regression Predicting z_bc_virality

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
*Interaction Effects (Style * Content)*						
Style:Charisma * Content:Congress	0.4158	0.504	0.826	0.409	−0.572	1.404
Style:Comedic * Content:Congress	−0.0495	0.386	−0.128	0.898	−0.807	0.708
Style:Communal * Content:Congress	−0.2165	0.399	−0.543	0.588	−0.999	0.566
Style:Critique * Content:Congress	−0.4043	0.446	−0.907	0.365	−1.279	0.470
Style:Event * Content:Congress	−0.1295	0.309	−0.419	0.675	−0.736	0.477
Style:Misc * Content:Congress	0.0451	0.479	0.094	0.925	−0.894	0.984
Style:Music * Content:Congress	0.0819	0.295	0.278	0.781	−0.496	0.660
Style:Vlog * Content:Congress	0.0478	0.345	0.139	0.890	−0.628	0.724
Style:Charisma * Content:Indep.	0.1282	0.333	0.385	0.701	−0.526	0.782
Style:Comedic * Content:Indep.	−0.3571	0.292	−1.224	0.221	−0.929	0.215
Style:Communal * Content:Indep.	0.4962	0.510	0.972	0.331	−0.505	1.497
Style:Critique * Content:Indep.	0.0847	0.318	0.266	0.790	−0.539	0.708
Style:Event * Content:Indep.	0.0238	0.280	0.085	0.932	−0.525	0.573
Style:Misc * Content:Indep.	−0.3807	0.311	−1.224	0.221	−0.991	0.230
Style:Music * Content:Indep.	0.0317	0.263	0.121	0.904	−0.484	0.547
Style:Vlog * Content:Indep.	−0.0524	0.314	−0.167	0.867	−0.668	0.563
Style:Charisma * Content:Maoist	−1.3742	0.612	−2.244	0.025	−2.575	−0.173
Style:Comedic * Content:Maoist	0.2530	0.536	0.472	0.637	−0.799	1.305
Style:Communal * Content:Maoist	−0.3835	0.497	−0.772	0.440	−1.357	0.590
Style:Critique * Content:Maoist	−0.7434	0.642	−1.159	0.247	−2.002	0.515
Style:Event * Content:Maoist	−1.0263	0.421	−2.439	0.015	−1.852	−0.201
Style:Misc * Content:Maoist	−1.0744	0.632	−1.701	0.089	−2.313	0.165
Style:Music * Content:Maoist	−0.7965	0.370	−2.153	0.032	−1.522	−0.071
Style:Vlog * Content:Maoist	−1.3151	0.533	−2.469	0.014	−2.360	−0.270
Style:Charisma * Content:UML	0.2157	0.432	0.499	0.618	−0.632	1.064
Style:Comedic * Content:UML	0.1083	0.374	0.289	0.772	−0.626	0.843
Style:Communal * Content:UML	0.0336	0.333	0.101	0.920	−0.621	0.688
Style:Critique * Content:UML	0.6429	0.459	1.401	0.161	−0.257	1.543
Style:Event * Content:UML	0.1307	0.304	0.431	0.667	−0.465	0.726
Style:Misc * Content:UML	0.1340	0.393	0.341	0.733	−0.636	0.904
Style:Music * Content:UML	0.2014	0.269	0.748	0.455	−0.327	0.729
Style:Vlog * Content:UML	0.0074	0.338	0.022	0.983	−0.656	0.671

Notes: This table reports the cleaned interaction-model sensitivity analysis after dropping four videos with invalid style code values. Reference category for Style is 'Civic Awareness'. Reference category for Content is 'Election'. Style abbreviations: Charisma = Charisma/Leadership, Critique = Critique/Frustration, Event = Event Coverage, Misc = Miscellaneous, Music = Music/Entertainment, Vlog = Personal Vlog/Family. Content abbreviations: Indep. = Independent. Standard errors are non-robust. The model includes 1526 observations. The interaction specification is more weakly conditioned than the reduced additive model and should be interpreted cautiously.

Table 9.12: Video Diversification Labels and Counts

Video Diversification ID	Video Diversification Labels	Count
0.0	0	2246
10083.0	0	277
10071.0	Lip-sync, Lip-sync, Performance	183
10075.0	Random Shoot, Others	67
10080.0	Finger Dance & Basic Dance, Singing & Dancing, Performance	20
10062.0	Cars, Trucks & Motorcycles, Auto & Vehicle, Lifestyle	19
10017.0	Babies, Family, Family & Relationship	17
10045.0	Movies & TV works, Entertainment Culture, Culture & Entertainment	14
10046.0	Music, Entertainment Culture, Culture & Entertainment	14
10014.0	Diary & VLOG, Daily Life, Lifestyle	9
10088.0	Celebrity Clips & Variety Show, Entertainment Culture, Culture & Entertainment	6
10056.0	Video Games, Games, Entertainment	6
1030.0	Lip-sync	5
10040.0	Cooking, Food & Drink, Lifestyle	5
10052.0	Comics & Cartoon, Anime, Anime & Comics	5
10024.0	Social News, Society, Society	4
10020.0	Farm Animals, Animals, Nature	3
10029.0	Outfit, Outfit, Beauty & Style	3
10003.0	Comedy, Performance	3
10009.0	Singing & Instruments, Singing & Dancing, Performance	3
10073.0	Selfies, Daily Life, Lifestyle	3
10070.0	Work & Jobs, Daily Life, Lifestyle	2
10081.0	Street Interviews & Social Experiments, Society, Society	2
10012.0	Romance, Relationship, Family & Relationship	2
10059.0	Traditional Sports, Sports, Sport & Outdoor	2
10058.0	Extreme Sports, Sports, Sport & Outdoor	2
10028.0	Nail Art, Beauty & Care, Beauty & Style	2
10063.0	Auto & Vehicle, Auto & Vehicle	1
1001.0	Scripted Drama	1
10019.0	Pets, Animals, Nature	1
10027.0	Beauty & Care, Beauty & Style	1
10087.0	Social Issues, Society, Society	1
10025.0	Hair, Beauty & Care, Beauty & Style	1
10095.0	Software & APPs, Technology, Culture & Entertainment	1

Table 9.13: Top Content Themes for `video_diversification_id = 10083.0`

Content Theme	**Count**
Speech/Explanatory	140
Independent (स्वतन्त्र) Support/Song	49
General Election Song	28
Funny	18
Congress (काँग्रेस/रुख) Support/Song	14
Maoist (हँसिया/हतौडा/Sickle) Support/Song	13
UML (सूर्य/एमाले) Support/Song	7
Others/Unclassified	2
TikTok Trends	2
Instrumental/Background Music	1
Sad Song	1
Nepali Patriotic Song	1
Nepali Folk/Traditional Song	1

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AI Disclosure

In accordance with the Hertie School's AI Guidelines (February 2023), I declare that I have used AI tools such as ChatGPT, Grok, and Google AI Studio for assistance in coding and thesis preparation. These tools were utilized in the following manner:

1. Code generation and debugging
2. Drafting and structuring certain sections of the text
3. Assistance with LaTeX formatting
4. Help with language refinement and clarity

I confirm that all AI-generated content has been carefully reviewed, edited, and verified by me, and I take full responsibility for all content, including any factual errors or inaccuracies that may exist in the final document. The use of these tools has been in accordance with the Hertie School's policy on academic integrity, with all significant contributions properly acknowledged.

Statement of Authorship

I hereby confirm and certify that this master thesis is my own work. All ideas and language of others are acknowledged in the text. All references and verbatim extracts are properly quoted and all other sources of information are specifically and clearly designated. I confirm that the digital copy of the master thesis that I submitted on 28th April 2025 is identical to the printed version I submitted to the Examination Office on 29th April 2025.

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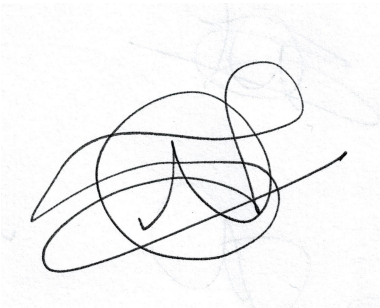
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